

EXP - 10 Database with VIEW and INDEX

AIM: To create view and index on the given relation.

DESCRIPTION:

VIEW: MySQL has supported database views since version 5+. In MySQL, almost features of views conform to the SQL: 2003 standard. MySQL processes query against the views in two ways:

1. In a first way, MySQL creates a temporary table based on the view definition
2. Statement and executes the incoming query on this temporary table.
3. In a second way, MySQL combines the incoming query with the query defined the view into one query and executes the combined query.

SYNTAX- VIEW:

```
CREATE [ALGORITHM = {MERGE | TEMPTABLE | UNDEFINED}]  
VIEW [database_name].[view_name]  
AS [SELECT statement]
```

INDEX:

A database index, or just index, helps speed up the retrieval of data from tables. When you query data from a table, first MySQL checks if the indexes exist, then MySQL uses the indexes to select exact physical corresponding rows of the table instead of scanning the whole table..

SYNTAX- INDEX:

```
CREATE [UNIQUE|FULLTEXT|SPATIAL] INDEX index_name  
USING [BTREE | HASH | RTREE]  
ON table_name (column_name [(length)] [ASC | DESC],...)
```

Questions on View:

1. Create a view with name 'v1' using employees1 table which holds the value of employee_id and salary of employee.
2. Do the insert and delete records from v1 table.

```
mysql> SELECT * FROM employee;
```

employee_id	first_name	last_name	device_serial	salary
101	Rahul	Sharma	DS1001	45000.00
102	Priya	Patel	DS1002	52000.00
103	Amit	Verma	DS1003	48000.00
104	Neha	Singh	DS1004	55000.00
105	Rohan	Gupta	DS1005	47000.00
106	Anjali	Mehta	DS1006	60000.00
107	Vikas	Kumar	DS1007	53000.00
108	Sneha	Reddy	DS1008	51000.00
109	Arjun	Nair	DS1009	62000.00
110	Pooja	Joshi	DS1010	49000.00
111	Karan	Malhotra	DS1011	57000.00
112	Meera	Iyer	DS1012	46000.00
113	Suresh	Das	DS1013	58000.00
114	Divya	Mishra	DS1014	54000.00
115	Nikhil	Chauhan	DS1015	50000.00
116	Kavya	Rao	DS1016	61000.00
117	Manoj	Yadav	DS1017	43000.00
118	Ishita	Saxena	DS1018	59000.00
119	Deepak	Bose	DS1019	56000.00
120	Ritika	Agarwal	DS1020	65000.00

```
20 rows in set (0.03 sec)
```

```
mysql> desc employee;
```

Field	Type	Null	Key	Default	Extra
employee_id	int	NO	PRI	NULL	
first_name	varchar(50)	YES		NULL	
last_name	varchar(50)	YES		NULL	
device_serial	varchar(20)	YES		NULL	
salary	decimal(10,2)	YES		NULL	

```
5 rows in set (0.10 sec)
```

```
mysql> CREATE VIEW v1 AS
-> SELECT employee_id, salary
-> FROM employee;
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> SELECT * FROM v1;
+-----+-----+
| employee_id | salary |
+-----+-----+
| 101         | 45000.00 |
| 102         | 52000.00 |
| 103         | 48000.00 |
| 104         | 55000.00 |
| 105         | 47000.00 |
| 106         | 60000.00 |
| 107         | 53000.00 |
| 108         | 51000.00 |
| 109         | 62000.00 |
| 110         | 49000.00 |
| 111         | 57000.00 |
| 112         | 46000.00 |
| 113         | 58000.00 |
| 114         | 54000.00 |
| 115         | 50000.00 |
| 116         | 61000.00 |
| 117         | 43000.00 |
| 118         | 59000.00 |
| 119         | 56000.00 |
| 120         | 65000.00 |
+-----+-----+
20 rows in set (0.01 sec)
```

```
mysql> DELETE FROM v1
-> WHERE employee_id = 121;
Query OK, 0 rows affected (0.05 sec)
```

```
mysql> SELECT * FROM v1;
+-----+-----+
| employee_id | salary |
+-----+-----+
| 101         | 45000.00 |
| 102         | 52000.00 |
| 103         | 48000.00 |
| 104         | 55000.00 |
| 105         | 47000.00 |
| 106         | 60000.00 |
| 107         | 53000.00 |
| 108         | 51000.00 |
| 109         | 62000.00 |
| 110         | 49000.00 |
| 111         | 57000.00 |
| 112         | 46000.00 |
| 113         | 58000.00 |
| 114         | 54000.00 |
| 115         | 50000.00 |
| 116         | 61000.00 |
| 117         | 43000.00 |
| 118         | 59000.00 |
| 119         | 56000.00 |
| 120         | 65000.00 |
+-----+-----+
20 rows in set (0.01 sec)
```

1. Create index1 for 'salary' attribute from employees1 relation and list the first name of the employees whose salary is above 145000 and explain the working principle of indexing and then drop the index1.
2. Create index1 for 'employee_id' attribute and display the first name of an employee whose employee id is 10 and explain the working principle of index1.

```
mysql> CREATE INDEX index1
      -> ON employee(salary);
Query OK, 0 rows affected (0.20 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> SELECT first_name
      -> FROM employee
      -> WHERE salary > 145000;
Empty set (0.01 sec)

mysql> UPDATE employee
      -> SET salary = 150000
      -> WHERE employee_id = 120;
Query OK, 1 row affected (0.05 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT first_name
      -> FROM employee
      -> WHERE salary > 145000;
+-----+
| first_name |
+-----+
| Ritika     |
+-----+
1 row in set (0.00 sec)
```